

Week 05 - Component Test Case Set

ScamGuard - Component / Unit Testing (Documentation Revision v2 Final)

Project	ScamGuard - Scam Image Detection Application
Activity	Code & Logic Reading Clinic
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Requirement baseline	SRS v1.1, branch <code>main</code> , 2026-09-12
Code baseline	<code>main</code> at 66bc9e4a
Document status	Revision Candidate v2 Final
Original submission	v1, tag <code>w05-submission-v1</code> , merge <code>ca354a8</code>

Revision note. v2 Final improves structure, traceability, terminology and readability only. It does not rewrite the original v1 submission history and does not change the frozen test basis, raw evidence or the 18 recorded test results.

Canonical requirement source: [Panuwat-ta/project - Document/srs](#)
Source-code repository: [Panuwat-ta/project](#)

1. Executive Summary

Week 05 verifies selected high-risk components of ScamGuard against the current canonical SRS v1.1. The scope covers Visual Analysis, Risk Score Calculation, Registration and Consent Management. Verification uses deterministic unit/component execution, a reproducible handler probe with fake DB, and explicit static contract inspection where an endpoint or output contract is absent.

The result set contains 18 cases: 7 Pass, 9 Fail, 1 Not Ready and 1 Needs Clarification. A failed case is retained as valid V&V evidence; no missing implementation is converted to Pass.

Result	Count	Interpretation
Pass	7	Verified behavior aligns with the baseline
Fail	9	Current implementation does not align with the baseline
Not Ready	1	Required component contract is not available for a valid acceptance test
Needs Clarification	1	Requirement example conflicts with the same baseline's algorithm semantics
Total	18	Selected Week 05 component cases

2. Test Basis, Scope and Strategy

2.1 Frozen test basis

- Canonical SRS: v1.1 dated 2026-09-12, from `Document/srs` on `main`
- SRS SHA-256:
`99bb8d5050fa54195b02869f1815e954fdd43ad0997e732b35426c70947cd40a`
- Code baseline: `Panuwat-ta/project` , branch `main` , commit `66bc9e4a`
- W03/W04 are historical v1.0 evidence only and do not override SRS v1.1

2.2 Selected scope

Scope	Requirement	Component	Verification focus
C01	FR-ANALYSIS-02	<code>onnx_worker.py</code>	forgery / AI-Gen / visual output contract
C02	FR-ANALYSIS-04	<code>risk_calculator.py</code>	Hybrid max+bonus and grade boundaries
C03	FR-AUTH-01	auth schema + handler	registration response and mandatory System Consent
C04	FR-PDPA-01	consent model + route inventory	consent lifecycle and profile/API contracts

2.3 Verification techniques

- Example-based tests for SRS-provided risk-score examples
- Boundary checks around Low/Medium/High transitions
- Negative handler probe for mandatory System Consent
- Static source/schema/route inspection for explicit output/API contracts
- Test-only fake DB and environment values; no production secret or database is used

Out of scope: model Accuracy/F1/mDice, GPU performance, full System/UAT journeys, production-service availability and unstated retroactive research-data rules. These concerns are handed to later V&V activities instead of being inferred in Week 05.

3. Component Test Cases

3.1 Visual Analysis - FR-ANALYSIS-02

ID	AC	Verification	Expected	Actual evidence	Result
CT-01	AC-1	Contract inspection	Separate <code>forgeries_confidence</code> 0-100 traceable to forgery signal	No <code>forgeries_confidence</code> ; worker exposes <code>visual_risk_score</code> from SegFormer max probability	Fail
CT-02	AC-2	Contract inspection	Separate <code>ai_gen_confidence</code> from an AI-Gen detector	<code>ai_gen_prob</code> is the same SegFormer max probability; no separate AI-Gen output	Fail
CT-03	AC-3	Readiness check	Visual-score behavior can be tested from canonical independent signals	Required separate signal contracts are not available on main	Not Ready

FR-ANALYSIS-02 AC-5 (GPU inference <= 10 seconds) is deferred to NFR testing because no valid performance evidence is produced by this component activity.

3.2 Risk Calculation - FR-ANALYSIS-04

ID	AC	Input / technique	Expected	Actual evidence	Result
CT-04	AC-1	(50,85,0) example	90, High, primary=visual	90 high visual True	Pass
CT-05	AC-2	(100,100,100) example	capped 100, High	100 high visual True	Pass
CT-06	AC-3	(10,0,0)	10, Low	10 low textual False	Pass
CT-07	AC-4	(30,0,0)	30, Low	30 low textual False	Pass
CT-08	AC-5	(55,0,0)	55, Medium	55 medium textual False	Pass
CT-09	AC-6	(80,0,0)	80, High	80 high textual False	Pass
CT-10	AC-7	consistency analysis of stated 65 total / 85 visual	High by Visual Override	Hybrid max uses visual in base, so total cannot be 65 when visual is 85	Needs Clarification

Supporting boundary probe: 39 -> Low, 40 -> Medium, 69 -> Medium, 70 -> High.

3.3 Registration - FR-AUTH-01

ID	AC	Verification	Expected	Actual evidence	Result
CT-11	AC-1	Handler probe + schema inspection	Response includes <code>id</code> , <code>full_name</code> , <code>email</code> , <code>role</code> , <code>status</code> , <code>created_at</code> ; no <code>password</code> / <code>hash</code>	Keys are <code>email</code> , <code>full_name</code> , <code>id</code> , <code>message</code> , <code>role</code> ; <code>status</code> and <code>created_at</code> absent	Fail
CT-12	AC-5	Negative handler probe: System Consent=false	HTTP 400 and no user/consent persistence	Handler returns <code>UserResponse</code> and fake DB receives <code>User + ConsentLog(false,false)</code>	Fail

CT-13	AC-1 subcondition	Handler probe: system=true, research=false	Optional Research Consent must not block registration	Registration succeeds and persists research=false	Pass
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3.4 Consent Management - FR-PDPA-01

ID	AC	Verification	Expected	Actual evidence	Result
CT-14	AC-2	Model contract inspection	Event/log model supports consent_type, is_granted and separate records	Current model stores one row with two booleans	Fail
CT-15	AC-3	Route inventory	PUT /consent/research updates research consent and records update	Endpoint not present on main	Fail
CT-16	AC-4	Route contract inspection	GET /users/me returns current-user profile	Profile GET is /api/v1/auth/me ; /api/v1/users/me is DELETE	Fail
CT-17	AC-5	Route inventory	GET /consent/logs returns consent history	Endpoint not present on main	Fail
CT-18	AC-3/5	Model contract inspection	Consent history/update supports updated_at	Current ConsentLog has created_at only	Fail

4. Evidence and Reproducibility

4.1 Evidence types

This artifact distinguishes executed tests, executed component probes and static contract inspection. Static inspection is used only where the SRS defines an explicit field/endpoint and the implementation contract can be verified directly; it is not presented as an end-to-end runtime test.

Evidence	Description	Supports
E05-baseline.txt	Code commit, SRS version and SRS SHA-256	Frozen test basis
E05-pytest-component.txt	Existing risk-calculator unit execution: 1 passed in 0.03s	Risk logic supporting evidence
E05-component-probe.txt	Risk examples/boundaries, auth handler behavior, route and visual contract inventory	CT-01 to CT-18 as mapped in work evidence
component_probe.py	Reproducible probe source using fake DB and test-only environment	Re-execution / audit

4.2 Evidence integrity

The test/probe execution used an isolated `origin/main` archive rather than the active product working branch.

No `.env`, API key, token, production password or production database content is included in this artifact.

No claim is made for model Accuracy/F1/mDice, GPU inference time, end-to-end system pass, stakeholder approval or independent peer-review approval because those claims are not supported by Week 05 evidence.

5. Findings and Handoff

ID	Finding	Status	Next action / handoff
F01	Visual forgery/AI-Gen signals are not separated as required	Open	Fix output contract/detector then re-test before System/UAT
F02	AC-7 example 65 total / 85 visual is inconsistent with Hybrid max+bonus semantics	Needs Clarification	Clarify SRS through controlled change; do not guess algorithm intent
F03	Registration response misses <code>status</code> and <code>created_at</code>	Open	Fix response schema/handler and add contract test
F04	System Consent=false is accepted and persisted	Open	Add guard before persistence and re-test negative case
F05	Consent persistence differs from event/log SRS contract	Open	Reconcile schema/migration before integration testing
F06	Research-consent update/log endpoints are missing	Open	Implement endpoints and component/API tests
F07	<code>/users/me</code> profile route differs between SRS and code	Open	Select canonical route then synchronize SRS/code/test
F08	W03/W04 used SRS v1.0	Closed for W05	Preserve as history; use v1.1 as current authority

Handoff. F01/F05/F06/F07 feed Week 06 integration risk; F01/F03/F04/F06/F07 feed Week 07 System/UAT after fixes; performance/model-quality concerns feed Week 11; unresolved product findings may become formal defects in Week 14.

6. AI Use and Review Integrity

AI was used to organize test cases, compare source contracts with the canonical SRS, execute local read-only verification through Remote Desktop Commander, and draft/report findings. Expected results were checked against SRS v1.1; actual results come from stored execution output or explicit source/schema/route inspection.

No test result, screenshot, stakeholder decision, peer-review approval or product metric was fabricated.

Review status

The original v1 artifact was merged through PR #5 at commit `ca354a8` and tagged `w05-submission-v1`. GitHub showed no recorded independent PR review at verification time, and the reviewer decision fields remain pending in `work/05-peer-review.md`.

Therefore this v2 Final document is a **Revision Candidate**. It may improve the quality of the Week 05 documentation without claiming that the independent peer-review gate has already passed.

Evidence package. Detailed scope, 18 test cases, raw evidence, findings, peer-review record and AI declaration are stored under `WEEKS/week-05/work/`. Canonical requirement and source repositories are linked on the cover page.